



Carbon Emissions Report 2025





Introduction

Kumaraguru Institutions is a multi-campus educational institution in Tamil Nadu, South India, offering undergraduate, postgraduate, and research programs in Engineering, Technology, Management, Agriculture, Liberal Arts, and Science. Guided by its motto, “Character is Life,” the institution is committed to academic excellence, holistic development, and nurturing responsible individuals with integrity and purpose. Through experiential learning, innovative research, and strong community engagement, Kumaraguru fosters a culture of inquiry and social responsibility, empowering students to address real-world challenges and contribute meaningfully to society.



Microcosm is a long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance, resource efficiency, and responsible environmental stewardship.

The program brings together students, faculty, and staff to work on projects and practices that enhance sustainability across the campus ecosystem from energy and water management to waste minimization and biodiversity conservation.

As part of its sustainability journey, Kumaraguru Institutions has undertaken the development of a Greenhouse Gas (GHG) emissions inventory in accordance with ISO 14064-1 to quantify emissions from institutional operations, and support structured initiatives for reducing its carbon footprint.

Highlights



EMISSION PROFILE

Total GHG
Emissions

1912.76
tCO2e

Emissions (DG +
Transport)

389.1
tCO2e

Emissions
(Electricity)

1364.7
tCO2e

Emissions (LPG -
Canteen)

158.96
tCO2e



ENERGY INDICATORS

Total Energy
Consumption

52,50,515
Kwh

Grid Energy
Consumption

1877147
Kwh

Renewable Energy
Consumption

3373368
Kwh

Renewable
Energy Share



Highlights



TRANSPORT INDICATORS (FLEET)

Total Fuel
Consumption

126034
Litres

Total Petrol
Consumption

8954.11
Litres

Total Diesel
Consumption

117080
Litres

Total Fleet
Emissions

337.6
tCO2e



RESOURCE EFFICIENCY INDICATORS

Total Waste
Generated

48.8
Tonnes

Total Recyclable
Waste

43.47
Tonnes

Total Organic
Waste

3.54
Tonnes

Others

1.74
Tonnes

Highlights



CLIMATE ACTION INDICATORS

Avoided
Emissions

3086.8
tCO₂e

Per Capita
Emissions

0.273
tCO₂e/6991 people



Purpose of the Report

This report presents the Greenhouse Gas (GHG) emissions inventory for Kumaraguru Institutions Institution for the reporting period January 1, 2025 to December 31, 2025, compiled in accordance with ISO 14064-1:2018. Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals. The objective of this inventory is to measure and disclose the institution's GHG emissions, and provide reliable data to support future emissions reduction strategies.

Organizational Boundary

The organizational boundary for the Greenhouse Gas (GHG) inventory has been defined using the operational control approach as prescribed under ISO 14064-1:2018. Under this approach, all facilities and operations over which Kumaraguru Institutions, located in Coimbatore, has operational control are included within the boundary of this inventory.

Operational control refers to the authority of the institution to introduce and implement operating policies at the facility or activity level. Based on this principle, the following campus operations are included:

- Academic and administrative buildings
- Laboratories and research centers
- Library and IT infrastructure facilities
- Student hostels and dining facilities (where managed by the institution)
- Utility infrastructure such as diesel generator sets, electrical distribution systems, and HVAC systems
- Institution-owned vehicles (if applicable)
- Common infrastructure such as lighting, water pumping, and campus services

Reporting Boundary

The GHG inventory covers Scope 1 and Scope 2 emissions

Scope 1 includes emissions from sources that are owned or operationally controlled by the institution. The following emission sources are included:

- Diesel consumption in backup generator sets
- Fuel consumption in institution-owned vehicles

Scope 2 includes indirect emissions resulting from the generation of purchased electricity consumed within campus operations.

The primary Scope 2 source considered is:

- Purchased grid electricity consumed across academic blocks, laboratories, hostels, administrative buildings, and campus utilities.

Reporting Period

This GHG inventory covers the reporting period from January 2025 to December 2025. The institution intends to update the inventory periodically to track emissions performance and improvement initiatives over time.

GHG Quantification Approach

The Greenhouse Gas (GHG) emissions inventory for Kumaraguru Institutions has been developed in accordance with the requirements of ISO 14064-1:2018 using an activity-based quantification approach. Emissions were calculated by multiplying activity data (e.g., fuel consumption, electricity consumption, refrigerant usage) with appropriate emission factors derived from recognized national and international sources.

The quantification approach follows guidance from the Intergovernmental Panel on Climate Change and the GHG Protocol to ensure transparency, completeness, consistency, and accuracy.

GWP (Global Warming Potential)

All greenhouse gases were converted into carbon dioxide equivalent (CO₂e) using Global Warming Potential (GWP) values from the IPCC Sixth Assessment framework (or latest available dataset used consistently across the inventory).

The primary gases considered include:

- Carbon dioxide (CO₂)
- Methane (CH₄)
- Nitrous oxide (N₂O)

Activity Data Collection

Activity data were collected through a structured process to ensure accuracy and completeness across all emission sources. Primary data were obtained wherever possible, including metered energy consumption, fuel usage records, or internal monitoring systems. When primary data was not available conservation averages were used as assumptions

Emission Factor

Emission factors used for the quantification of greenhouse gas (GHG) emissions for Kumaraguru Institutions were selected based on internationally accepted methodologies and nationally relevant reference sources to ensure transparency, consistency, and conservativeness in the inventory.

For fuel combustion sources (diesel and petrol), default emission factors were adopted from the guidelines published by the Intergovernmental Panel on Climate Change (IPCC), as supplier-specific emission factors were not available for the fuels consumed within the institutional operations.

The emission factors include the three primary greenhouse gases associated with stationary and mobile combustion:

- Carbon dioxide (CO₂)
- Methane (CH₄)
- Nitrous oxide (N₂O)

All gases were converted into carbon dioxide equivalent (CO₂e) using Global Warming Potential (GWP) values.

Diesel Parameters Used

- NCV: 43 MJ/kg
- Density: 0.845 kg/litre
- Energy conversion: 36.335 MJ/litre
- **CO₂e Emission Factor: 2.701 kgCO₂e per litre**

Petrol Parameters Used

- NCV: 44.3 MJ/kg
- Density: 0.775 kg/litre
- Energy conversion: 34.3325 MJ/litre
- **CO₂e Emission Factor: 2.388 kgCO₂e per litre**

Supplier-specific emission factors were not available for the fuels consumed during the reporting period. Therefore:

- IPCC default emission factors were applied following the Tier 1 approach.
- Fuel density values were sourced from BS-VI fuel specifications to reflect regionally relevant fuel characteristics.
- Conservative assumptions were applied to avoid underestimation of emissions.
- DEFRA 2025 emissions were used for LPG estimation

These assumptions are consistent with ISO 14064-1 principles of accuracy, completeness, and conservativeness.

Emission Factor

For the quantification of Scope 2 emissions associated with purchased electricity consumption at Kumaraguru Institutions, the grid emission factor published by the Central Electricity Authority (CEA), Government of India, was used.

Grid Emission Factor Used: 0.727 kgCO₂e/kWh

Exclusions

Fugitive emissions, primarily associated with refrigerant leakage from air-conditioning and refrigeration systems, were identified as a potentially relevant category. However, during the reporting period, consolidated and reliable data on refrigerant inventory, leakage rates, or refill quantities were not available across all facilities.

Scope 3

For Kumaraguru Institutions, Scope 3 emissions were assessed at a screening level during the reporting period to identify relevant indirect emission sources associated with institutional activities. Scope 3 emissions include indirect emissions that occur from sources not owned or directly controlled by the institution but are a consequence of its operations.

Potential Scope 3 categories identified for an educational campus include:

- Student and staff commuting
- Business travel
- Purchased goods and services (e.g., laboratory consumables, office materials)
- Waste generated in operations
- Outsourced services
- Capital goods and infrastructure development

However, during the current reporting cycle, Scope 3 emissions have not been quantitatively calculated due to the unavailability of consolidated and verifiable activity data across these categories. Data for Scope 3 sources is currently dispersed across multiple departments and external stakeholders, and standardized tracking mechanisms are not yet fully established.

Given the absence of consistent datasets and the need to maintain accuracy and reliability in accordance with ISO 14064-1 principles, Scope 3 emissions have been excluded from quantitative reporting for this baseline inventory.

Transport Emissions (Scope 1)

337.6 tCO₂e

MONTH	PETROL (Litres)	EMISSIONS (tCO ₂ e)	DIESEL (Litres)	EMISSIONS (tCO ₂ e)
JANUARY	746	1.7814	9757	26.3536
FEBRUARY	483.5	1.1545	7230	19.5282
MARCH	493	1.1772	9063	24.4791
APRIL	252	0.6017	5771	15.5874
MAY	440	1.0507	6435	17.3809
JUNE	414	0.9886	9184	24.8059
JULY	853	2.0369	8955	24.1874
AUGUST	908	2.1683	9184	24.8059
SEPTEMBER	1123	2.6817	8588	23.1961
OCTOBER	976.22	2.3312	15060	40.6770
NOVEMBER	1124.6	2.6855	13629	36.8119
DECEMBER	1140.79	2.7242	14224	38.4190



DG Set Emissions (Scope 1)

51.5

 tCO₂e

MONTH	DIESEL (Litres)	EMISSIONS (tCO ₂ e)
JANUARY	1588.62	4.2908
FEBRUARY	1319.34	3.5635
MARCH	274.23	0.7406
APRIL	1273.8	3.4405
MAY	1628.22	4.3978
JUNE	951.39	2.5697
JULY	214.17	0.5784
AUGUST	1849.32	4.9950
SEPTEMBER	2959.11	7.9925
OCTOBER	3368.64	9.0986
NOVEMBER	2162.82	5.8417
DECEMBER	1475.43	3.9851



Electricity Emissions (Scope 2)

1364.7 tCO_{2e}

Renewable Energy
Consumption

3373368 kWh

GRID Energy
Consumption

1877147 kWh

Total
Consumption

5250515 kWh

MONTH	GRID HT CONNECTION		GRID COMMERCIAL CONNECTION		GRID TEMPORARY CONNECTION	
	ELECTRICITY (KWh)	EMISSIONS (tCO _{2e})	ELECTRICITY (KWh)	EMISSIONS (tCO _{2e})	ELECTRICITY (KWh)	EMISSIONS (tCO _{2e})
JANUARY	155222	112.846	1023	0.743	0	0
FEBRUARY	194516	141.413	862	0.626	0	0
MARCH	261397	190.035	1119	0.813	0	0
APRIL	139071	101.104	1140	0.828	0	0
MAY	1321	0.960367	1001	0.727	0	0
JUNE	690	0.50163	1023	0.743	0	0
JULY	5934	4.314	1128	0.820	428	0.311
AUGUST	424410	308.546	1094	0.795	416	0.302
SEPTEMBER	450216	327.307	1027	0.746	389	0.282
OCTOBER	76958	55.948	1000	0.727	346	0.251
NOVEMBER	105958	77.0314	963	0.700	331	0.240
DECEMBER	46972	34.148	899	0.653	293	0.213

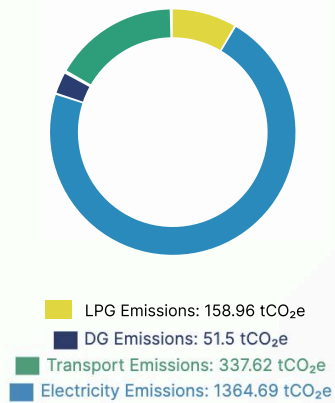
LPG Emissions Canteen (Scope 1)

158.96 tCO₂e

MONTH	LPG (Litres)	EMISSIONS (tCO ₂ e)
JANUARY	8778	13.67
FEBRUARY	8816	13.73
MARCH	8322	12.96
APRIL	6897	10.74
MAY	8930	13.91
JUNE	8854	13.79
JULY	9006	14.02
AUGUST	7942	12.37
SEPTEMBER	8417	13.11
OCTOBER	9595	14.94
NOVEMBER	7429	11.57
DECEMBER	9101	14.17



Source Distribution



Transport Sources

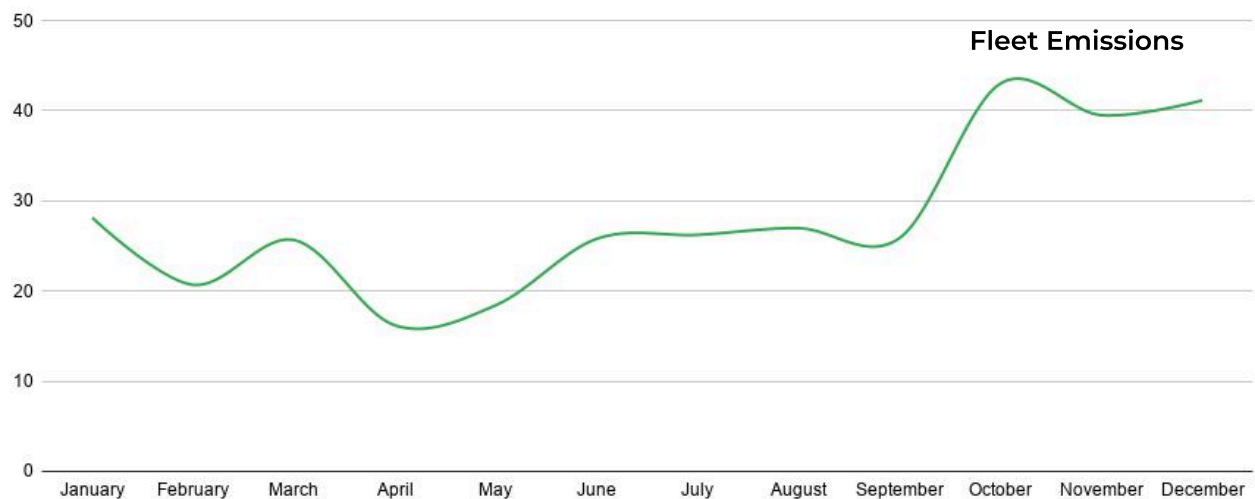
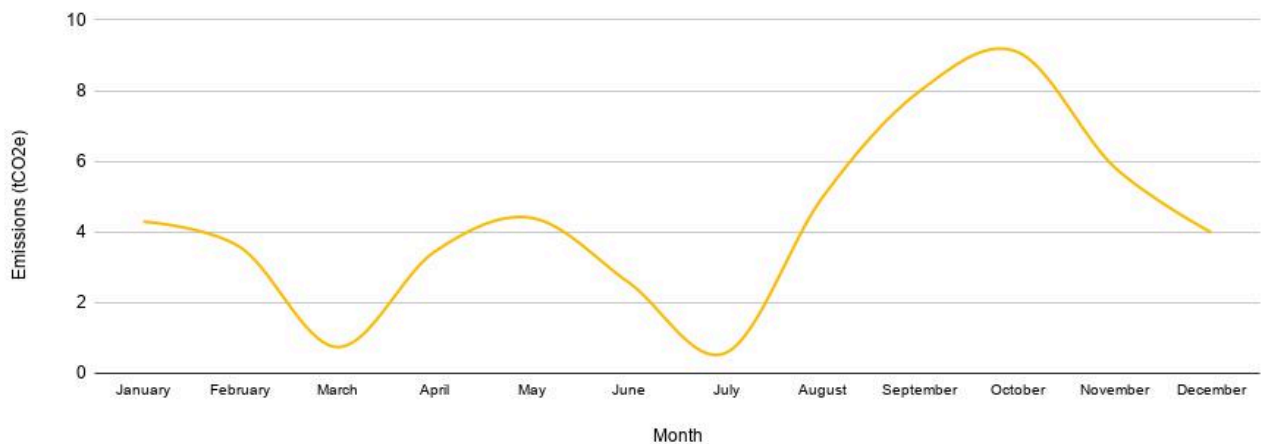


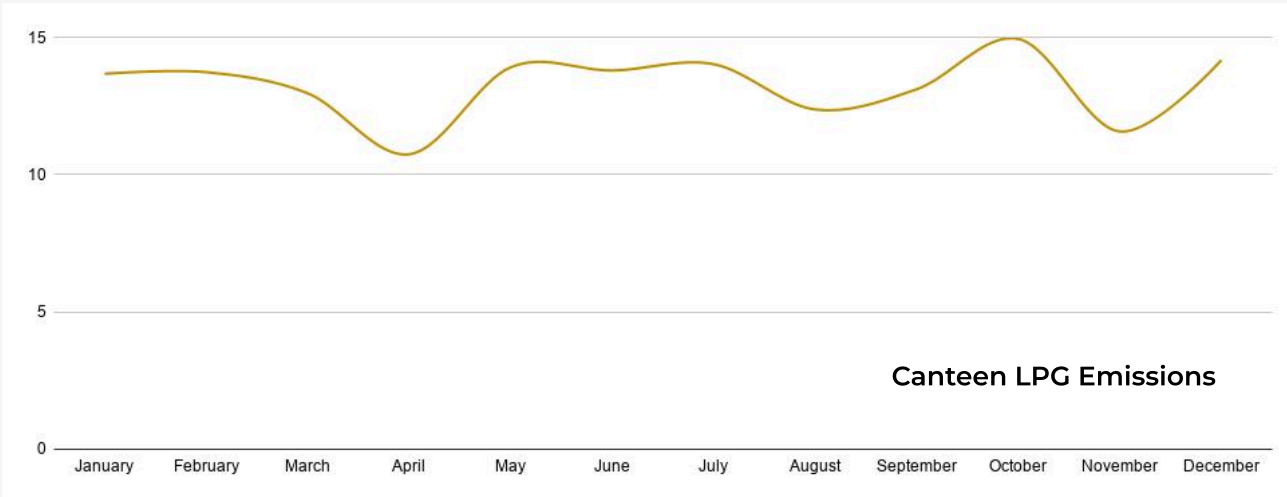
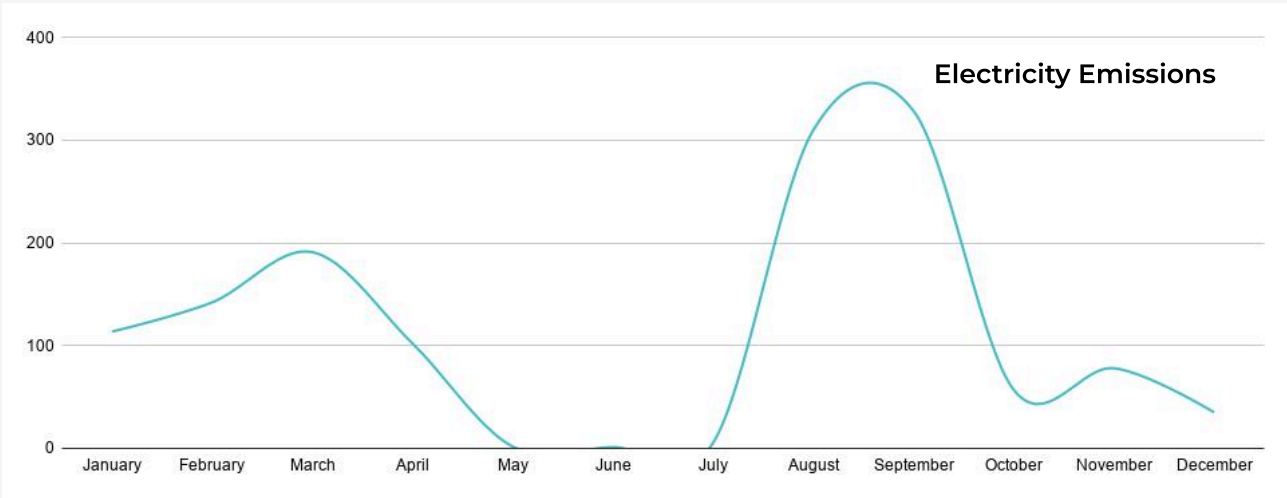
Highest Emission Month
Sep — 362.21 tCO₂e



Lowest Emission Month
May — 24.52 tCO₂e

Emissions (tCO₂e) vs. Month





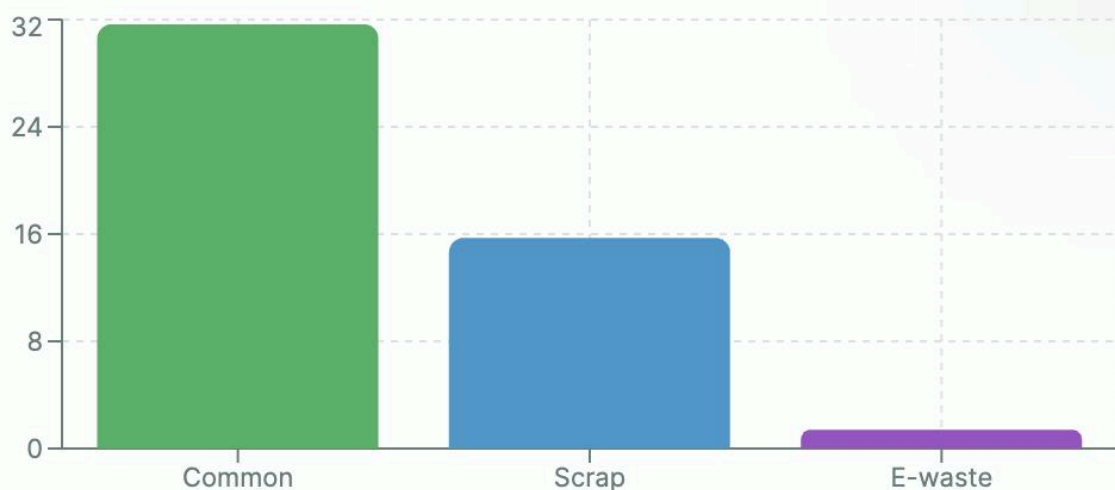
	2024	2025	% Change	
Total Emissions	2448.935	1912.76	-21.89%	
Scope 1 Emissions	304.605	548.06	+79.92%	
Scope 2 Emissions	2144.330	1364.7	-36.35%	

Waste

The institution operates a Resource Recovery Park, where waste collected from academic blocks, hostels, and campus facilities is segregated and processed through recycling, composting, upcycling, incineration, and scientific disposal methods. A structured process involving daily collection, transportation, and segregation enables effective handling of multiple waste streams such as organic waste, paper, plastics, e-waste, metals, glass, and sanitary waste.

Category	Item	Quantity (Tonnes)
Common	Wet Items	1.938
Common	PET	0.812
Common	Plastic/Black Plastic	3.627
Common	LDPE	0.409
Common	White/Colour Papers/Newspapers	14.079
Common	Cardboards/PP Boards	10.787
Scrap	PVC Items/Doors/Windows	3.922
Scrap	Iron	6.356
Scrap	SS Materials	0.516
Scrap	Aluminium/Light Weight	2.965
Scrap	Coconut Shell	1.605
Scrap	Others	0.346
E-waste	E-waste Total	1.400

Waste Distribution (Tonnes)



Consolidated GHG Summary

ISO 14064	GHG Protocol	Emissions (tCO ₂ e)
Category 1 - Direct Emissions	Scope 1 Emissions	548.06
	Stationary Combustion	210.46
	Mobile Combustion	337.6
	Process Emissions	Not Calculated
	Fugitive Emissions	Not Calculated
	Biogenic Emissions	Not Applicable
Category 2 - Indirect Emissions from imported energy	Scope 2 Emissions	1364.7
	Purchased electricity (location-based)	1364.7
Category 3 - Indirect Emissions from Transportation	Scope 3 Emissions	0
	Category 4 - Upstream Transportation & Distribution	Not Calculated
	Category 6 - Business Travel	Not Calculated
	Category 7 - Employee Commute	Not Calculated
	Category 9 - Downstream Transportation & Distribution	Not Calculated
Category 4 - Indirect Emissions from products used by organization	Category 1 - Purchased Goods & Services	Not Calculated
	Category 2 - Capital Goods	Not Calculated
	Category 3 - Fuel and Energy related activities	Not Calculated
	Category 5 - Waste generated in operations	Not Calculated
	Category 8 - Upstream leased assets	Not Calculated
Category 5 - Indirect Emissions from use of organisation's products	Category 10 - Processing of Sold products	Not Calculated
	Category 11 - Use of Sold products	Not Calculated
	Category 12 - End-of-Life treatment of sold products	Not Calculated
	Category 13 - Downstream leased assets	Not Calculated
	Category 14 - Franchises	Not Calculated
	Category 15 - Investments	Not Calculated
Total Emissions (Scope 1 + 2 + 3)		1912.76

Avoided Emissions

As part of its energy management and sustainability initiatives, Kumaraguru Institutions has incorporated renewable energy sources within its campus electricity consumption mix. Renewable electricity reduces dependence on grid-based electricity, which is primarily generated from fossil fuel sources, thereby contributing to avoided greenhouse gas (GHG) emissions.

Avoided emissions represent the emissions that would have occurred if the equivalent quantity of electricity had been sourced entirely from the conventional grid.

The Following activities have helped us reduce our emissions

- On-site solar power generation installed within the campus
- Third-party solar power procurement through renewable energy supply arrangements
- Use of Solar Water heater

Avoided emissions were estimated using the grid emission factor published by the Central Electricity Authority (CEA), which represents the carbon intensity of electricity generation in India.

The following approach was applied:

Avoided Emissions (tCO₂e) = Renewable Electricity Consumption (kWh) × Grid Emission Factor (kgCO₂/kWh) ÷ 1000

Avoided Emission Initiatives

Renewable Energy

On-site Solar Generation

143.8 tCO₂e

Renewable Energy

Third-party Solar PPA

2308.6 tCO₂e

Renewable Energy

Solar Water Heater

545.3 tCO₂e

Offset

Green Cover

89.1 tCO₂e

Conclusion

This Greenhouse Gas (GHG) inventory report for Kumaraguru Institutions has been prepared in accordance with the requirements of ISO 14064-1:2018 to quantify emissions associated with institutional operations for the defined reporting period. The inventory provides a structured and transparent assessment of greenhouse gas emissions across the identified organizational and operational boundaries.

This inventory enhances the understanding of emission sources within campus operations and provides a foundation for tracking environmental performance and identifying opportunities for energy efficiency and emission reduction. The institution intends to progressively improve data quality, expand emission coverage, and integrate climate considerations into operational decision-making.

REPORT PREPARED BY

Beyond Sustainability is a purpose-driven consultancy dedicated to helping businesses transition beyond conventional sustainability practices towards truly regenerative and resilient operations. Rooted in the philosophy that sustainability is not just about minimizing harm but about actively contributing to environmental and social well-being, we work with organizations to embed sustainability at the core of their strategies.

